

CURRICULUM VITAE

Seung-Mo Lee, PhD

PERSONAL INFORMATION

Name: Seung-Mo Lee
 Gender: Male
 Nationality: South Korea
 Affiliation: **R&D Staff, Senior Scientist**
 Department of Nanomechanics
 Korea Institute of Machinery & Material (KIMM)
 Sinseongno 104, Yuseong-gu, Daejeon, 305-343
 South Korea

Professor
 in Korea University of Science and Technology
 (UST), Daejeon, South Korea



Contact Email: sm.lee@kimm.re.kr; wisethinking@ust.ac.kr; immrlee@gmail.com
 Tel: +82 42 868 7659, +82 10 7449 8552
 Fax: +82 42 868 7176

APPOINTMENTS

May / 2016 ~ **Visiting Scientist**
 Nov/2016 Max Planck Institute of Colloids and Interfaces, Potsdam, Germany
 In the Department of Biomaterials (Prof. Dr. Peter Fratzl)

 Sept / 2012 ~ **Assistant/Associate/Full Professor**
 Present University of Science and Technology (UST), Daejeon, South Korea

 Mar / 2011 ~ **Senior Scientist**
 Present Nano-Mechanical System Research Division
 in Korea Institute of Machinery & Material (KIMM), South Korea

 Jan / 2010 ~ **Research Associate**
 April / 2011 Max Planck-Institute of Microstructure Physics, Germany
 (under the supervision of Prof. Dr. Peter Fratzl and Prof. Dr. Mato Knez)

 Mar / 2007 ~ **Research Assistant**
 Dec / 2009 Max-Planck-Institut für Mikrostrukturphysik, Germany
 (under the supervision of Prof. Dr. Ulrich Gösele, **passed away on Nov 8, 2009**)

- Sept / 2006 ~ **Research Assistant**
 Feb/ 2007 Lehr- und Forschungsgebiet Konstruktion und Entwicklung von Mikrosystemen (KEmikro), Rheinisch-Westfälische Technische Hochschule (RWTH) Aachen, Germany
- Mar / 2005 ~ **Research Assistant**
 May / 2006 Advanced Materials Processing Lab, Pohang University of Science and Technology (POSTECH), South Korea
 (under the supervision of Prof. Dr. Tai Hun Kwon, **passed away on Mar 20, 2010**)

EDUCATION

- Dec 2009 **Ph.D.** in Applied Science (Physics Major)
 Albert-Ludwigs Universität Freiburg, Germany
 ► Thesis: *Atomic Layer Deposition on Biological Matter*
 ○ Advisor: Prof. Dr. Ulrich Gösele (**passed away on Nov 8, 2009**)
- Feb 2005 **M.S.** in Mechanical Engineering
 Pohang University of Science and Technology (POSTECH), South Korea
 ► Thesis: *Fabrication of Hydrophobic Films Using a Plant Leaf in Nature*
 ○ Advisor: Prof. Dr. Tai Hun Kwon (**passed away on Mar 20, 2010**)
- Feb 2003 **B.S.** in Mechanical Engineering
 Inha University, South Korea
 ► Thesis: *A Design of Vibration Absorber*
 ○ Advisor: Prof. Dr. Usik Lee

TRAINING AND EDUCATIONS

- June / 2006 *German Language Course*
 Sept/ 2007 Carl Duisberg Centren, Köln, Germany

HONORS AND AWARDS

- Deutscher Akademischer Austausch Dienst (DAAD) Scholarship for PhD Program

ADDITIONAL CAPABILITIES

- Language: English, German, Korean
 Sport: Tennis, Badminton, Swimming
 Extracurricular activity: Calligraphy, Hiking, Acoustic Guitar, Photography

RESEARCH INTERESTS

- Mechanical Behavior and Microstructure of Bulk, Thin Film, and Composite Materials
- Preparation and Characterization of Composite Materials: Relationships between

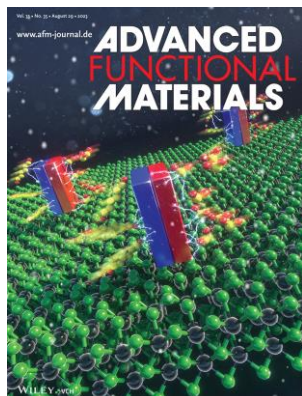
Microstructure and Properties of Composite Materials, Structure and Properties of Interfaces in Composite Materials

- Biologically Inspired Materials
- Prediction and Modeling of Properties in Composite Materials.
- Experimental Mechanics
- Fracture Characteristics of Composites and Biological Materials
- Atomic Layer Deposition (ALD) Process
- Graphene Growth, Characterization, and Applications
- Synthesis and Application of graphene oxide, metal-organic framework, Mxene
- Energy storage materials

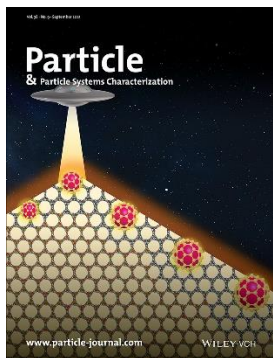
PUBLICATIONS

A. Peer-Reviewed Scientific Journal Articles (* Corresponding Author)

- [79] Winda Devina, Iyan Subiyanto, Dao Thi Dung, Seong Ok Han, Hyung Chul Yoon, Do Van Lam, **Seung-Mo Lee***, Hyunuk Kim
Chem. Phys. Rev. **4**, 041308 (2023)
Rapid synthesis of nanomaterials by solvent-free laser irradiation for energy storage and conversion
- [78] Viet Phuong Nguyen, Jae-Hyun Kim, **Seung-Mo Lee***
Mater. Today Energy **38**, 101451 (2023)
Strategies of Boosting Synergistic Adsorption-Catalysis Effect for High-Performance Lithium-Sulfur Batteries
- [77] Do Van Lam, Dao Thi Dung, Jae-Hyun Kim, **Seung-Mo Lee***
ACS Appl. Energy Mater. **6**, 5368-5377 (2023)
Metal–Organic Framework-Derived NiO@C as a Host for MnO₂ Cathode of Stable Zinc-Ion Batteries
- [76] Dao Thi Dung, Euijin Roh, Sanghyeon Ji, Jong Min Yuk, Jae-Hyun Kim, Hyunuk Kim, **Seung-Mo Lee***
Nanoscale **15**, 1794-1805 (2023)
Ni/Co/Co₃O₄@C nanorods derived from a MOF@MOF hybrid for efficient overall water splitting
- [75] Viet Phuong Nguyen, Ji Su Park, Hyung Cheoul Shim, Jong Min Yuk, Jae-Hyun Kim, Duckjong Kim, **Seung-Mo Lee***
Adv. Funct. Mater. **33**, 2303503 (2023)
“Accelerated Sulfur Evolution Reactions by TiS₂/TiO₂@MXene Host for High-Volumetric-Energy-Density Lithium–Sulfur Batteries”
- [74] Do Van Lam, Dao Thi Dung, Jae-Hyun Kim, Hyunuk Kim, **Seung-Mo Lee***
Chem. Eng. J. **437**, 135237 (2022)
Laser-induced sulfurization of nickel-based metal-organic frameworks for highly stable phase-engineered energy materials



- [73] Viet Phuong Nguyen, Il Hwan Kim, Hyung Cheoul Shim, Ji Su Park, Jong Min Yuk, Jae-Hyun Kim, Duckjong Kim, **Seung-Mo Lee***
Energy Stor. Mater. **46**, 542-552 (2022)
 Porous carbon textile decorated with VC/V₂O₃-X hybrid nanoparticles: Dual-functional host for flexible Li-S full batteries
- [72] Viet Phuong Nguyen, Ji Su Park, Jong Min Yuk, Minsub Oh, Jae-Hyun Kim, **Seung-Mo Lee***
J. Mater. Chem. A **10**, 13428-13438 (2022)
 “Boosted Zn ²⁺ storage performance of hydrated vanadium oxide by defect and heterostructure”
- [71] Yeongcheol Park, Jaehoon Kim, Changho Kim, **Seung-Mo Lee**, Chul Kang, Jae Hun Seol
Int. J. Heat Mass Transf. **188**, 122632 (2022)
 “In situ and operando thermal characterization in aqueous electric double layer capacitors using the 3 ω hot-wire method”
- [70] Uyen Nhat Trieu Nguyen, Khai Hoang Do, Bongkyun Jang, Kyung-Shik Kim, Jae-Hyun Kim, **Seung-Mo Lee***
 “Always-on photocatalytic antibacterial facemask with mini UV-LED array”
Mater. Today Sustain. **18**, 100117(2022)
- [69] Woosung Kim, Wonseok Lee, **Seung-Mo Lee**, Duckjong Kim, Jinsung Park
Nanotechnology **33**, 175702 (2022)
 “Enhanced thermoelectric performance of defect engineered monolayer graphene”
- [68] Do Van Lam, Dao Thi Dung, Euijin Roh, Jae-Hyun Kim, Hyunuk Kim, **Seung-Mo Lee***
Adv. Mater. Interfaces **8**, 2101599(2021)
 “Metal-organic framework-templated graphitic carbon confining MnO/Mn₃O₄ nanoparticles via direct laser printing for electrocatalysis and supercapacitor”

- [67] Seunghwan Choy, Hoang Tran Bui, **Seung-Mo Lee**, Wooyul Kim, Dong Soo Hwang
Carbohydr. Polym. **271**, 118413 (2021)
“Photocatalytic exoskeleton: chitin nanofiber for retrievable and sustainable TiO₂ carriers for the decomposition of various pollutants”
- [66] Viet Phuong Nguyen, Duckjong Kim, **Seung-Mo Lee***
ACS omega **6**, 29054-29059 (2021)
“Tuning the thermal conductivity of the amorphous PAA polymer via vapor-phase infiltration”
- [65] Muhammad Sohail, Van-Toan Nguyen, Quang-Tung Ngo, Anto Jeffery, Ho-Suk Choi, Namgee Jung, Jae-Hyun Kim, Hyunuk Kim, **Seung-Mo Lee***
Chem. Eng. J. **421**, 127731 (2021)
“Laser-scribed ultrasmall nanoparticles with unary and binary phases”
- [64] Do Van Lam, Van-Toan Nguyen, Euijin Roh, Quang-Tung Ngo, Wanuk Choi, Jae-Hyun Kim, Hyunuk Kim, Ho-Suk Choi, **Seung-Mo Lee***
Part. Part. Syst. Charact. **38**, 2100119 (2021)
“Laser-Induced graphitic carbon with ultrasmall nickel nanoparticles for efficient overall water splitting” (Cover)
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- [63] Do Van Lam, Uyen Nhat Trieu Nguyen, Euijin Roh, Wanuk Choi, Jae-Hyun Kim, Hyunuk Kim, **Seung-Mo Lee***
Small **17**, 2100670 (2021)
“Graphitic carbon with MnO/Mn₇C₃ prepared by laser-scribing of MOF for versatile supercapacitor electrodes”
- [62] Uyen Nhat Trieu Nguyen, Hyung Cheoul Shim, **Seung-Mo Lee***
Renew. Energy **171**, 116-123 (2021)
“Leaf-derived porous carbon synthesized by carbothermic reduction”

- [61] Viet Phuong Nguyen, Mikyung Lim, Kyung-Shik Kim, Jae-Hyun Kim, Ji Su Park, Jong Min Yuk, **Seung-Mo Lee***
J. Mater. Chem. A **9**, 10739-10746 (2021)
“Drastically increased electrical and thermal conductivities of Pt-infiltrated MXenes”
- [60] Viet Phuong Nguyen, Jin Yoo, Ju Young Lee, Justin J Chung, Jeong Ho Hwang, Youngmee Jung, **Seung-Mo Lee***
ACS Appl. Mater. Interfaces **12**, 43501-43512 (2020)
“Enhanced mechanical stability and biodegradability of Ti-infiltrated polylactide”
- [59] Do Van Lam, Muhammad Sohail, Jae-Hyun Kim, Hak Joo Lee, Seong Ok Han, Jonghwa Shin, Duckjong Kim, Hyunuk Kim, **Seung-Mo Lee***
ACS Appl. Mater. Interfaces **12**, 39154-39162 (2020)
“Laser synthesis of MOF-derived Ni@ carbon for high-performance pseudocapacitors”
- [58] Yongjune Kim, Duckjong Kim, Sang-Hun Lee, Minah Seo, Hyun-June Jung, Boyoung Kang, **Seung-Mo Lee**, Hak-Joo Lee
Opt. Express **28**, 17143-17152 (2020)
“Single-layer metamaterial bolometer for sensitive detection of low-power terahertz waves at room temperature”
- [57] Nguyen Minh Hieu, Truong Thi Hien, Nguyen Duc Chinh, Nguyen Duc Quang, Nguyen Manh Hung, Cao Van Phuoc, **Seung-Mo Lee**, Jong-Ryul Jeong, Chunjoong Kim, Dojin Kim
Mater. Des. **191**, 108628 (2020)
“ZnTe-coated ZnO nanorods: hydrogen sulfide nano-sensor purely controlled by pn junction”
- [56] **Seung-Mo Lee***, Yong-Jin Park, Jae-Hyun Kim, Kyubock Lee
Appl. Surf. Sci. **512**, 145626 (2020)
“Effects of annealing on electrochemical performance in graphene/V2O5 supercapacitor”
- [55] Sejeong Won, Hyun-June Jung, Dasom Kim, Sang-Hun Lee, Hyeon-Don Kim, Kwang-Seop Kim, **Seung-Mo Lee**, Minah Seo, Dai-Sik Kim, Hak-Joo Lee, Jae-Hyun Kim
Carbon **158**, 505-512 (2020)
“Graphene-based crack lithography for high-throughput fabrication of terahertz metamaterials”

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- [54] Yong-Jin Park, Jae-Hyun Kim, Kyubock Lee, **Seung-Mo Lee***
Korean Chem. Eng. Res. **58**, 135-141 (2020)
“Fabrication of VOx/graphene composite using CO2 laser reduction and atomic layer deposition and its electrochemical performance”
- [53] Do Van Lam, Sejeong Won, Hyung Cheoul Shim, Jae-Hyun Kim, **Seung-Mo Lee***
Carbon, **153**, 257-264 (2019)
“Turning cotton into tough energy textile via metal oxide assisted carbonization”
- [52] Seunghwan Choy, Do Van Lam, **Seung-Mo Lee**, Dong Soo Hwang
ACS Appl. Mater. Interfaces **11**, 38440-38447 (2019)
“Prolonged biodegradation and improved mechanical stability of collagen via vapor-phase Ti stitching for long-term tissue regeneration”
- [51] **Seung-Mo Lee***, Yong-Jin Park, Jae-Hyun Kim
ACS Appl. Nano Mater. **2**, 3711-3717 (2019)
“Laser reduction of Zn-infiltrated multilayered graphene oxide as electrode materials for supercapacitors”
- [50] Nguyen Duc Chinh, Truong Thi Hien, Lam Do Van, Nguyen Minh Hieu, Nguyen Duc Quang, **Seung-Mo Lee**, Chunjoong Kim, Dojin Kim
Sens. Actuators B Chem. **281**, 262-272 (2019)
“Adsorption/desorption kinetics of nitric oxide on zinc oxide nano film sensor enhanced by light irradiation and gold-nanoparticles decoration”
- [49] Seunghwan Choy, Dongyeop X Oh, Seungwon Lee, Do Van Lam, Gihoon You, Jin-Soo Ahn, Seung-Woo Lee, Sang-Ho Jun, **Seung-Mo Lee***, Dong Soo Hwang
ACS Appl. Mater. Interfaces **11**, 9786-9793 (2019)
“Tough and immunosuppressive titanium-infiltrated exoskeleton matrices for long-term endoskeleton repair”
- 48] Dhruvajyoti Bhattacharjya, Chang-Hyeon Kim, Jae-Hyun Kim, In-Kyu You, Jung Bin In, **Seung-Mo Lee***
Appl. Surf. Sci. **462**, 353-361 (2018)
“Fast and controllable reduction of graphene oxide by low-cost CO2 laser for supercapacitor application”
- [47] Duckjong Kim, Woochang Kim, **Seung-Mo Lee**, Tengfei Luo, Choongman Moon, Jae-

Hyun Kim, Hak-Joo Lee, Jinsung Park, Joo-Hyoung Lee

Int. J. Therm. Sci. **133**, 266-272(2018)

“Paved phonon transport route in graphene by vapor phase process”

- [46] Tung Xuan Tran, Hayelin Choi, Cuong Huu Che, Ji Hwan Sul, In Gyoo Kim, **Seung-Mo Lee**, Jae-Hyun Kim, Jung Bin In

ACS Appl. Mater. Interfaces **10**, 39777-39784 (2018)

“Laser-induced reduction of graphene oxide by intensity-modulated line beam for supercapacitor applications”

- [45] Chang Hyun Kim, Jae-Hyun Kim, **Seung-Mo Lee***

ACS omega **3**, 7904-7910 (2018)

“Transformation of solid waste into nanoporous carbon via carbothermic reduction”

- [44] Jeong Hwan Lee, Su Ji Choi, Ji Hwan Kwon, **Seung-Mo Lee**, An Soon Kim, Hion Suck Baik, Sang Jung Ahn, Seong Gu Hong, Yong Ju Yun, Young Heon Kim

Phys. Chem. Chem. Phys **20**, 16276-16284 (2018)

“Abnormal elastic modulus behavior in a crystalline–amorphous core-shell nanowire system”

- [43] Do Van Lam, Hyung Cheoul Shim, Jae-Hyun Kim, Hak-Joo Lee, **Seung-Mo Lee***

Small **13**, 1702702 (2017)

“Carbon textile decorated with pseudocapacitive VC/VxOy for high-performance flexible supercapacitors”

- [42] Sang-Min Kim, Bongkyun Jang, Kyungmin Jo, Donghyuk Kim, Jihye Lee, Kyung-Shik Kim, **Seung-Mo Lee**, Hak-Joo Lee, Seung Min Han, Jae-Hyun Kim

2D Materials **4**, 2 (2017)

“Mechanics-driven patterning of CVD graphene for roll-based manufacturing process”

- [41] Ji-ung Choi, Duckjong Kim, **Seung-Mo Lee**, Junkyu Han, Hyemi So, Junghyo Nah & Jae Hun Seol

Int. J. Biol. Macromolec. **96**, 384–391 (2017)

“Effect of β -sheet crystals on the thermal conductivity of spider dragline silk”

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- [40] Do Van Lam, Kyungmin Jo, Chang-Hyun Kim, Jae-Hyun Kim, Hak-Joo Lee & **Seung-Mo Lee***
ACS Nano **10**, 11351–11359 (2016)
“Activated carbon textile via chemistry of metal extraction for supercapacitors”
- [39] Truong Thi Hien, Do Van Lam, Chunjoong Kim, Nguyen Minh Vuong, Nguyen Duc Quang, Dahye Kim, Nguyen Duc Chinh, Nguyen Minh Hieu, **Seung-Mo Lee** & Dojin Kim
J. Power Sources **336**, 367–375 (2016)
“Enhanced carrier collection efficiency in hierarchical nano-electrode for a high-performance photoelectrochemical cell”
- [38] Cheolmin Park, **Seung-Mo Lee** & Won Seok Chang
Phys. Chem. Chem. Phys. **18**, 26184–26191 (2016)
“Carrier transport behaviors depending on the two orthogonally directional energy bands in the ZnO nanofilm affected by oxygen plasma”
- [37] Do Van Lam, Kyungmin Jo, Chang-Hyun Kim, Sejeong Won, Hwangbo Yun, Jae-Hyun Kim, Hak-Joo Lee & **Seung-Mo Lee***
J. Mater. Chem. A **4**, 4082–4088 (2016)
“Calligraphic ink enabling washable conductive textile electrode for supercapacitor”
- [36] Min Young Na, **Seung-Mo Lee**, Do Hyang Kim & Hye Jung Chang
Applied Microscopy **45**, 23–31 (2015)
“Dark-field TEM imaging technique to visualize the local structure of two-dimensional material: graphene”
- [35] Tao Gong, Do Van Lam, Renlong Liu, Sejeong Won, Hwangbo Yun, Sanghyuk Kwon, Jinseon Kim, Ke Sun, Jae-Hyun Kim, **Seung-Mo Lee*** & Changgu Lee*
Adv. Funct. Mater. **25**, 3756–3763 (2015)
“Thickness dependence of the mechanical properties of free-standing graphene oxide papers”
- [34] Kyungmin Jo, Sangbong Lee, Sang-Min Kim, Jung Bin In, **Seung-Mo Lee**, Jae-Hyun Kim, Hak-Joo Lee & Kwang-Seop Kim
Chem. Mater. **27**, 3621–3627 (2015)
“Stacked bilayer graphene and redox-active interlayer for transparent and flexible high-performance supercapacitors”

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- [33] Hye-Mi So, Hyekyoung Choi, Hyung Cheoul Shim, **Seung-Mo Lee**, Sohee Jeong & Won Seok Chang
Appl. Phys. Lett. **106**, 093507 (2015)
“Atomic layer deposition effect on the electrical properties of Al₂O₃-passivated PbS quantum dot field-effect transistors”
- [32] **Seung-Mo Lee***, Sang-Min Kim, Min Young Na, Hye Jung Chang, Kwang-Seop Kim, Hyunung Yu, Hak-Joo Lee & Jae-Hyun Kim
Nano Res. **8**, 2082-2091 (2015)
“Materialization of strained CVD-graphene using thermal mismatch”
- [31] Do Van Lam, Tao Gong, Sejeong Won, Jae-Hyun Kim, Hak-Joo Lee, Changgu Lee & **Seung-Mo Lee***
Chem. Comm. **51**, 2671-2674 (2015)
“A robust and conductive metal-impregnated graphene oxide membrane selectively separating organic vapors”
- [30] **Seung-Mo Lee**, Jae-Hyun Kim & Jong-Hyun Ahn
Mater. Today **18**, 336-344 (2015)
“Graphene as flexible electronic materials: mechanical limitation by defect formation and efforts to overcome” (Invited Review Article)
- [29] Kyungmin Jo, Sang-Min Kim, **Seung-Mo Lee**, Jae-Hyun Kim, Hak-Joo Lee, Ki Soo Kim, Yong Duck Kwon & Kwang-Seop Kim
Carbon **82**, 168-175 (2015)
“One-step etching, doping, and adhesion-control process for graphene electrodes”
- [28] Youngji Cho, Jun-Mo Yang, Do Van Lam, **Seung-Mo Lee**, Jae-Hyun Kim, Kwan-Young Han, Jiho Chang,
Applied Microscopy **44**, 133-137 (2014)
"Transmission electron microscopy specimen preparation for layer-area graphene by a direct transfer method"
- [27] Gyeong-Man Kim, **Seung-Mo Lee**, Mato Knez & Paul Simon
Thin Solid Films **562**, 291-298 (2014)
“Single phase ZnO submicrotubes as a replica of electrospun polymer fiber template by atomic layer deposition”
- [26] Cheolmin Park, Hye-Mi So, Hyeon Jun Jeong, Mun Seok Jeong, Eckhard Pippel, Won

- Seok Chang & **Seung-Mo Lee***
ACS Appl. Mater. Interfaces **6**, 16243-16248 (2014)
"Hierarchically structured ZnO/Petal hybrid composites with tuned optoelectronic and mechanical properties"
- [25] **Seung-Mo Lee***, Eckhard Pippel, Oussama Moutanabbir, Jae-Hyun Kim, Hak-Joo Lee & Mato Knez
ACS Appl. Mater. Interfaces **6**, 16827-16834 (2014)
"In Situ Raman spectroscopic study of Al-Infiltrated spider dragline silk under tensile deformation"
- [24] Sejeong Won, Yun Hwangbo, Seoung-Ki Lee, Kyung-Shik Kim, Kwang-Seop Kim, **Seung-Mo Lee**, Hak-Joo Lee, Jong-Hyun Ahn, Jae-Hyun Kim & Soon-Bok Lee
Nanoscale **6**, 6057-6064 (2014)
"Double-layer CVD graphene as stretchable transparent electrodes"
- [23] Do Van Lam, Sang-Min Kim, Youngji Cho, Jae-Hyun Kim, Hak-Joo Lee, Jun-Mo Yang & **Seung-Mo Lee***
Nanoscale **6**, 5639-5644 (2014)
"Healing defective CVD-graphene through vapor phase treatment"
- [22] Sangmin Kim, Sang-Min Kim, Jae-Hyun Kim, Kwang-Seop Kim, Yun Hwangbo, Jong-Hyuk Yoon, Eun-Kyu Lee, Jaechul Ryu, Hak-Joo Lee, Seungmin Cho & **Seung-Mo Lee***
Nanoscale **6**, 4728-4734 (2014).
"Synthesis of CVD-graphene on rapidly heated copper foils"
- [21] Choong-Kwang Lee, Yun Hwangbo, Sang-Min Kim, Seoung-Ki Lee, **Seung-Mo Lee**, Seong-Su Kim, Kwang-Seop Kim, Hak-Joo Lee, Byung-Ik Choi, Chang-Kyu Song, Jong-Hyun Ahn* & Jae-Hyun Kim*
ACS Nano **25**, 2336-2344 (2014).
"Monatomic chemical-vapor-deposited graphene membranes bridge a half-millimeter-scale gap"
- [20] Yun Hwangbo, Choong-Kwang Lee, Jae-Hyun Kim, Sang-Min Kim, Kwang-Seop Kim, Bongkyun Jang, Hak-Joo Lee, Seung-Ki Lee, Seong-Su Kim, Jong-Hyun Ahn* & **Seung-Mo Lee***
Sci. Rep. **4**, 4439 (2014).
"Fracture characteristics of monolayer CVD-graphene"

- [19] **Seung-Mo Lee***, Kwang-Seop Kim, Eckhard Pippel, Sangmin Kim, Jae-Hyun Kim & Hak-Joo Lee

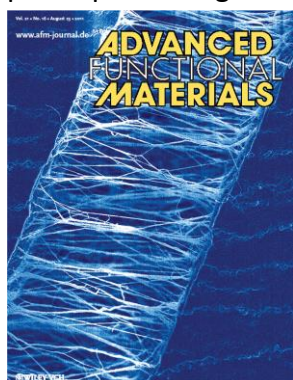
J. Phys. Chem. C **116**, 1781-2790 (2012)

“Facile route toward mechanically stable superhydrophobic copper using oxidation-reduction induced morphology changes”

- [18] **Seung-Mo Lee***, Vladislav Ischenko*, Eckhard Pippel, Admir Masic, Oussama Moutanabbir, Peter Fratzl & Mato Knez*

Adv. Funct. Mater. **21**, 3047–3055 (2011) (Cover)

“An alternative route towards metal-polymer hybrid materials prepared by vapor phase processing”



- [17] **Seung-Mo Lee***, Eckhard Pippel & Mato Knez*

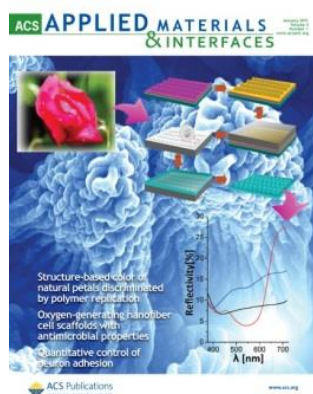
ChemPhysChem **12**, 791-798 (2011)

“Metal infiltration into biomaterials by ALD and CVD: A comparative study”

- [16] **Seung-Mo Lee***, Johannes Üpping, Andreas Bielawny & Mato Knez*

ACS Appl. Mater. Interfaces **3**, 30-34 (2011) (Cover)

“Structure-based color of natural petals discriminated by polymer replication”



- [15] Martin Otto*, Matthias Kroll, Thomas Käsebier, **Seung-Mo Lee**, Matti Putkonen, Roland Salzer, Paul T. Miclea & Ralf B. Wehrspohn

Adv. Mater. **22**, 5035-5038 (2010)

"Conformal transparent conducting oxides on black silicon"

- [14] **Seung-Mo Lee***, Eckhard Pippel, Oussama Moutanabbir, Ilja Gunkel, Thomas Thurn-Albrecht & Mato Knez

ACS Appl. Mater. Interfaces **2**, 2436-2441 (2010)

"Improved mechanical stability of dried collagen after metal infiltration"

- [13] Yong Qin,* Yunseok Kim, Lianbing Zhang, **Seung-Mo Lee**, Ren Bin Yang, AnLian Pan, Klaus Mathwig, Marin Alexe, Ulrich Gösele & Mato Knez

Small **6**, 910-914 (2010)

"Preparation and elastic properties of helical nanotubes obtained by ALD with carbon nanocoils as templates"

- [12] Johannes Üpping,* Andreas Bielawny, **Seung-Mo Lee**, Mato Knez, Reinhard Carius & Ralf B. Wehrspohn

Proceedings of SPIE, **7409**, 74090J (2009)

"Electric transport in 3D photonic crystal intermediate reflectors for micromorph thin-film tandem solar cells"

- [11] **Seung-Mo Lee***, Gregor Grass, Gyeong-Man Kim, Christian Dresbach, Lianbing Zhang, Ulrich Gösele & Mato Knez

Phys. Chem. Chem. Phys. **11**, 3608-3614 (2009)

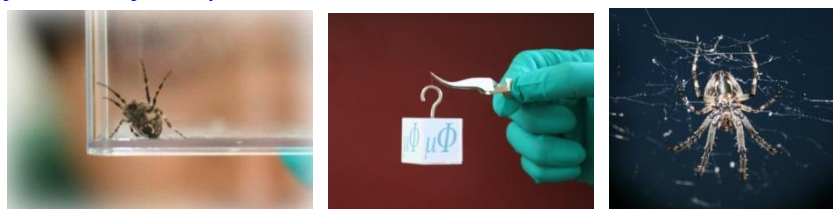
"Low-temperature ZnO atomic layer deposition on biotemplates: flexible photocatalytic ZnO structures from eggshell membranes"

- [10] **Seung-Mo Lee***, Eckhard Pippel, Ulrich Gösele, Christian Dresbach, Yong Qin, C. Vinod Chandran, Thomas Bräuniger, Gerd Hause & Mato Knez

Science **324**, 488-492 (2009)

"Greatly increased toughness of infiltrated spider silk"

- *Reported and highlighted in many newspapers, magazines, internet websites and journals of many countries*



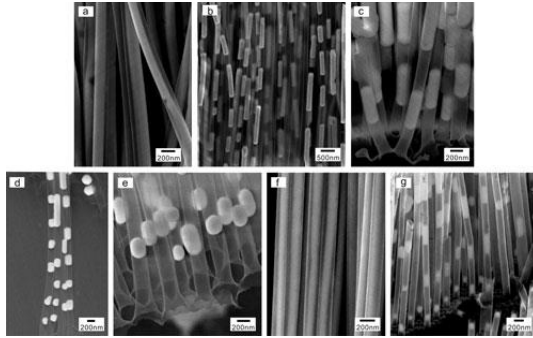
- [09] Andreas Bielawny,* Johannes Üpping, Paul T. Miclea, Ralf B. Wehrspohn, Carsten Rockstuhl, Falk Lederer, Marius Peters, Lorenz Steidl, Rudolf Zentel, **Seung-Mo Lee**,

Mato Knez, Andreas Lambertz & Reinhard Carius

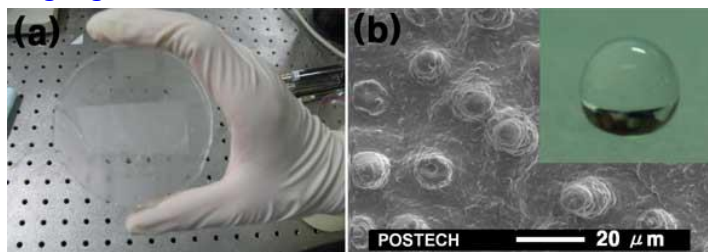
phys. stat. sol.(a) **205**, 2796-2810 (2008)

“3D photonic crystal intermediate reflector for micromorph thin-film tandem solar cell”

- [08] Dong Sik Kim, * **Seung-Mo Lee**, Roland Scholz, Mato Knez, Ulrich Gösele, Johannes Fallert, Heinz Kalt & Margit Zacharias
Appl. Phys. Lett. **93**, 103108 (2008)
“Synthesis and optical properties of ZnO and carbon nanotube based coaxial heterostructures”
- [07] Andreas Bielawny, * Paul T. Miclea, Ralf B. Wehrspohn, **Seung-Mo Lee**, Mato Knez, Carsten Rockstuhl, Marian Lisca, Falk L. Lederer & Reinhard. Carius
Proceedings of SPIE, **7002**, 700208 (2008)
“Three-dimensional photonic crystals as intermediate filter for thin-film tandem solar cells”
- [06] Yang Yang, * Dong Sik Kim, Roland Scholz, Mato Knez, **Seung-Mo Lee**, Ulrich Gösele & Margit Zacharias
Chem. Mater. **20**, 3487-3494 (2008)
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PATENTS (Domestic)

발명구분	발명의 명칭	출원번호	출원일	등록번호	등록일	주발명자	전체발명자
1 국내특허	대면적 연속면적 측정장치	2012-0021866	2012-03-02	1282983	2013-07-01	장봉균	장봉균,장봉균,김재현,김광섭,이승모,김경식,이혁주,최병익
2 국내특허	조소성 나노구조체 및 이의 제조방법	2011-0130706	2011-12-08	1337328	2014-03-06	이승모	이승모,김재현,이혁주
3 국내특허	그라핀 발광장치	2011-0130769	2011-12-08	1131062	2012-03-21	황보윤	황보윤,김상민,이승모,이혁주,김재현,김경식
4 국내특허	판상형 구조의 나노박막과 기판 사이의 신뢰성 향상 장치 및 방법	2012-0015041	2012-02-14	1173300	2012-08-06	김광섭	김광섭,장봉균,최현주,황보윤,이승모,김상민,김재현,이혁주,김경식
5 국내특허	판상형 구조의 나노박막과 기판 사이의 신뢰성 향상 장치	2012-0060650	2012-06-05	1241964	2013-03-05	김광섭	김광섭,장봉균,최현주,황보윤,이승모,김상민,김재현,이혁주,김경식
6 국내특허	플로팅 기판의 그라핀 연속 합성 장치	2012-0022102	2012-03-05	1168259	2012-07-18	김상민	김상민,이혁주,김재현,김경식,이승모,장봉균,황보윤,김광섭,최현주,오충석
7 국내특허	시트형 심플론관용 홀더 및 그것을 갖는 심플론관용 버퍼터	2012-0033126	2012-06-11	1206384	2012-11-23	이승모	이승모,김상민,김재현,이혁주,장봉균,김광섭,최현주,황보윤,김광성
8 국내특허	인덕션 히팅을 이용한 웨이퍼 단위의 그라핀 연속 합성 장치 및 방법	2012-0060777	2012-06-11	1206384	2012-11-23	김상민	김상민,이혁주,김재현,김경식,오충석,이승모,김광성,메이서알렉산더
9 국내특허	임프린트 기법을 이용한 나노박막 패턴 형성·전사 방법 및 장치	2012-0130742	2013-10-31	1579935	2015-12-17	이승모	이승모,도반람,김상민,김광섭,김재현,장봉균,김광성,이혁주
10 국내특허	그라핀의 결합 자유 방법 및 결합이 자유된 그라핀	2014-0039789	2014-04-03	1580252	2015-12-18	이승모	이승모,도반람,김상민,김광섭,김재현,장봉균,김광성,이혁주
11 국내특허	원지중 적층 슈퍼커패시터 및 이의 제조방법	2014-0135527	2014-10-08	1605819	2016-03-17	김재현	김재현,장봉균,이혁주,조경민,이승모,김경식,최병익,이상록
12 국내특허	그라핀 결합 적층장치 및 방법	2014-0093010	2014-07-23	1615117	2016-04-19	김정열	김정열,김상민,이승모,김광성
13 국내특허	절연막이 코팅된 발열필터	2014-0055070	2014-05-08	1517991	2015-04-29	김덕중	김덕중,김재현,이승모,우창수
14 국내특허	산화 그라핀 복합적층체 및 이의 제조방법	2014-0162266	2014-11-20	1580243	2015-12-18	이승모	이승모,조경민,김광섭,김상민,이승모,김재현,이혁주
15 국내특허	투영 그라핀 전극과 이의 제조방법, 및 이를 이용한 슈퍼커패시터	2015-0041332	2015-03-25	1740899	2017-05-23	조경민	조경민,김광섭,김상민,이승모,김재현,이혁주
16 국내특허	다중 균단을 가진 반도체 검사용 프로브 및 프로브 모듈	2015-0176176	2015-12-10	1640884	2016-07-13	김정열	김정열,이승모,김광성
17 국내특허	활성탄 탄소전극의 제조방법	2015-0164424	2015-11-24	1721493	2017-03-24	김창현	김창현,조경민,심형철,홍성진,김재현,이승모
18 국내특허	활성탄 탄소전극의 제조방법	2016-0023014	2016-02-26	1769413	2017-08-11	조경민	조경민,도반람,심형철,김광섭,김재현,이혁주,이승모
19 국내특허	적층형 슈퍼커패시터의 제조방법	2016-0022386	2016-02-25	1774253	2017-08-29	김재현	김재현,이승모,장봉균,유인규,홍성훈,김주미,김빛나
20 국내특허	세탁 가능한 유연 섬유 전극 및 이의 제조방법	2016-0061913	2016-02-05	1827321	2018-02-02	이승모	이승모,도반람,조경민,김재현
21 국내특허	박막형 샘플 보관용 홀더	2016-0052965	2016-04-29	1867312	2018-06-07	이승모	이승모,김재현
22 국내특허	계측회로 모듈을 이용한 선택적 전사 플러그	2016-0151058	2016-05-20	1729574	2017-04-18	김광섭	김광섭,김재현,장봉균,김재현,이승모,황보윤
23 국내특허	경도기판 캐리어 필름	2017-0041871	2017-03-31	1885158	2018-07-30	김광섭	김광섭,김재현,장봉균,김재현,이승모,황보윤
24 국내특허	버전도상 전적박막을 이용한 마이크로 소자 전사방법	2017-0011817	2017-01-25	1918106	2018-11-07	김재현	김광섭,황보윤,장봉균,김경식,김재현,이혁주,최병익,이승모
25 국내특허	불완전 캐리어 필름에 방법 및 이를 위한 리퍼어 장치	2017-0132250	2016-10-12	1866901	2018-06-05	김재현	김광섭,이재학,김재현,황보윤,장봉균,김경식,이혁주,최병익,이승모
26 국내특허	다중층 캐리어 필름을 이용한 소자 전사 방법과 이 방법을 이용하여 전자제품을	2016-0132250	2016-10-12	1831455	2017-12-21	장봉균	장봉균,김재현,장봉균,장봉균,김광섭,김경식,이혁주,최병익,이승모
27 국내특허	플러그로의 정렬편 및 제어장치	2016-0132250	2016-10-12	1921694	2018-11-19	이승모	이승모,김재현,황보윤,김경식,김광섭,이혁주,최병익,장봉균
28 국내특허	탄소 열원공을 이용한 활성탄 원 구조체의 제조 방법 및 상기 제조 방법에 의하여 제	2017-0035069	2017-11-07	2090028	2020-03-11	이승모	이승모,김재현,이혁주,황보윤,김광섭,김경식,장봉균
29 국내특허	유리기 하이드리드 상체전해질 다공성 지지체 및 이의 제조방법	2017-0147174	2017-12-11	2077049	2020-02-07	황보윤	황보윤,윤성욱,김경식,김재현,김광섭,이혁주,최병익,장봉균
30 국내특허	마이크로 소자 전사방법 및 마이크로 소자 전사장치	2017-0177206	2017-11-10	1963875	2019-03-25	이승모	이승모,김재현,이혁주,황보윤,김광섭,장봉균,김경식
31 국내특허	탄소 복합체 및 그 제조방법	2017-0149399	2017-12-21	2077049	2020-02-07	김광섭	김광섭,이재학,김재현,황보윤,윤성욱,김경식,장봉균,이승모,최병익
32 국내특허	마이크로 소자 어레이 전사방법	2017-0158348	2017-11-24	2012237	2019-08-13	김광섭	김광섭,이재학,김재현,황보윤,윤성욱,김경식,장봉균,이승모,최병익
33 국내특허	유리기 복합체 및 이의 제조방법	2018-0016975	2018-02-12	2129316	2020-06-26	이승모	이승모,김재현,김경식,황보윤,김광섭,장봉균,이혁주,문영은
34 국내특허	줄 히팅을 이용한 그라핀 제조장치 및 이의 제조방법	2018-0000253	2018-01-02	2129316	2020-06-26	정현준	정현준,김재현,문영은,이혁주,장봉균,김광섭,이승모,황보윤
35 국내특허	탄소 복합체 및 그 제조방법	2018-0080530	2018-07-11	2100736	2020-04-08	이승모	이승모,김재현,김광섭,황보윤,장봉균,김경식
36 국내특허	마이크로 소자의 국면 전사방법 및 마이크로 소자의 국면 전사장치	2018-0080375	2018-07-11	2108385	2020-04-29	장봉균	장봉균,김재현,김경식,이승모,정현준,이혁주
37 국내특허	소자 전사 제어기 가능한 사이트 및 이를 이용한 소자 간격 제어방법	2018-0130135	2018-10-29	2123419	2020-06-10	장봉균	장봉균,김재현,최병익,박현성,김광섭,이승모
38 국내특허	필름 전사 방법	2018-0130136	2018-10-29	2154811	2020-09-04	장봉균	장봉균,김재현,박현성,이승모,김광섭,최병익
39 국내특허	전사기판 용수를 위한 메타구조체를 이용한 온도 감응형 센서	2018-0166688	2018-12-20	2146381	2020-08-13	김용준	김용준,김덕중,정현준,장보영,이혁주,이승모
40 국내특허	개선했을 공극과 레진 전착을 위한 지르코니아의 표면결정 방법과 활용	2019-0024920	2019-03-04	2230701	2021-03-16	최소환	이승모,황동수,최소환,이도환,안진수,주성원,전상호
41 국내특허	신축성계수 구간을 가지는 메타구조체 및 재료 탄성계수 구간을 가지는 메타구조체	2019-0069765	2019-06-13			장봉균	장봉균,김재현,임세형,이혁주,이승모,김광섭
42 국내특허	신축성기판 및 이를 이용한 소자 간격 제어방법	2020-0112696	2020-09-03			장봉균	장봉균,김재현,김광섭,김용성,이승모,최병익,이재학
43 국내특허	전사용 스템프 제조방법	2020-0135836	2020-10-29			장봉균	장봉균,김재현,김광섭,김용성,이승모,최병익,이재학
44 국내특허	탄화 메트릭스-입자 복합체 및 이의 제조방법	2020-0113918	2020-09-07	2252431	2021-05-10	이승모	이승모,장봉균,김재현
45 국내특허	상시 구동 가능한 광촉매 마스크 및 이의 제조방법	2021-0035444	2021-03-18			이승모	이승모,임미정,김경식,김재현
46 국내특허	금속이 집속된 메트릭 복합체 및 이의 제조방법 및 이를 포함하는 열전 소자	2021-0078009	2021-06-16			이승모	이승모,임미정,김경식,김재현
47 국내특허	3차원 구조를 갖는 탄소/금속 나노 입자 복합 구조체의 제조방법, 3차원 구조를 갖는	2021-0041485	2021-03-30			임현의	임현의,박승철,오선홍,여선주,이보연,김영수,김성기,권만우,이승모
48 국내특허	분자 레벨도판팅 장치 및 이를 이용한 도판팅 방법	2021-0041491	2021-03-30			김재현	김재현,장봉균,임형준,김현준,이승모
49 국내특허	플라기 기반 간의 평행도 검출장치 및 검출장치를 이용한 평행도 정렬방법	2021-0177588	2021-12-13			김재현	김재현,장봉균,임형준,김현준,이승모

PATENTS (International)

발명구분	국가	발명의 명칭	출원번호	출원일	등록번호	등록일	주발명자	전제발명자
1 해외특허	PCT	투명 그래픽 디스플레이의 제조방법 및 이를 이용한 슈퍼캐시터	PCT/KR2015/	2015-08-27			조경민	조경민,김광섭,김상민,이승모,김재현,이학주
2 해외특허	PCT	적층형 슈퍼캐시터의 제조방법	PCT/KR2017/	2017-02-24			김재현	김재현,이승모,장봉균,유인규,홍성훈,김주미,김빛나
3 해외특허	PCT	계층화 메타물질 제작용 선택적 전사 롤 스텝프	PCT/KR2016/	2016-11-29			김광섭	김광섭,김찬,장봉균,김재현,이승모,황보윤
4 해외특허	PCT	경도가변 캐리어 필름	PCT/KR2017/	2017-12-19			김광섭	김광섭,황보윤,장봉균,김경식,김재현,이학주,최병익,이승모
5 해외특허	PCT	다중형 캐리어 필름 및 이를 이용한 소자 전사 방법과 이 방법을 이용	PCT/KR2017/	2017-10-12			김재현	김재현,황보윤,장봉균,김광섭,김경식,이학주,최병익,이승모
6 해외특허	PCT	롤 스텝프의 정속변위 제어장치	PCT/KR2017/	2017-10-12			장봉균	장봉균,김재현,김경식,황보윤,김광섭,최병익,이학주,이승모
7 해외특허	PCT	마이크로 소자 전사방법 및 마이크로 소자 전사장치	PCT/KR2018/	2018-12-20			황보윤	황보윤,윤성욱,김경식,김재현,김광섭,이학주,최병익,박현성,김정엽,이승모,장봉균
8 해외특허	PCT	탄소 복합체 제조방법, 그에 의해 제조된 탄소 복합체, 및 탄소 복합체	PCT/KR2018/	2018-11-09			이승모	이승모,김재현,이학주,황보윤,김광섭,장봉균,김경식
9 해외특허	PCT	유무기 복합체 및 이의 제조방법	PCT/KR2018/	2018-12-18			이승모	이승모,김재현,김경식,황보윤,김광섭,장봉균,이학주,문명운
10 해외특허	PCT	마이크로 소자의 국면 전사방법 및 마이크로 소자의 국면 전사장치	PCT/KR2019/	2019-06-21			장봉균	장봉균,김재현,김경식,이승모,정현준,이학주
11 해외특허	PCT	소자 간격 제어가 가능한 시트 및 이를 이용한 소자 간격 제어방법	PCT/KR2019/	2019-10-28			장봉균	장봉균,김재현,최병익,박현성,김광섭,이승모
12 해외특허	PCT	필름 전사 방법	PCT/KR2019/	2019-10-28			장봉균	장봉균,김재현,박현성,이승모,김광섭,최병익
13 해외특허	PCT	전자기파 흡수를 위한 메타구조체 및 이를 이용한 온도 감응형 센서	PCT/KR2019/	2019-12-18			김용준	김용준,김덕중,정현준,김보영,이학주,이승모
14 해외특허	PCT	신축성 기판 및 이를 이용한 소자 간격 제어방법	PCT/KR2019/	2019-12-18			장봉균	장봉균,원세장,김재현,이학주,이승모,김광섭,박아현
15 해외특허	PCT	적층형 슈퍼캐시터의 제조방법	PCT/KR2021/	2021-08-31			김재현	김재현,이승모,장봉균,유인규,홍성훈,김주미,김빛나
16 해외특허	미국	선택적 전사 롤 스텝프	16/079746	2018-08-24	10418188	2019-09-17	김광섭	김광섭,김찬,장봉균,김재현,이승모,황보윤
17 해외특허	미국	마이크로 소자의 국면 전사방법 및 마이크로 소자의 국면 전사장치	15/733942	2020-12-01	10744750	2020-08-18	장봉균	장봉균,김재현,김경식,이승모,정현준,이학주
18 해외특허	미국	필름 전사 방법	17/289324	2021-04-28			장봉균	장봉균,김재현,박현성,이승모,김광섭,최병익
19 해외특허	일본	다중형 캐리어 필름 및 이를 이용한 소자 전사 방법과 이 방법을 이용	2019-517749	2018-12-10	6807454	2020-12-09	김재현	김재현,황보윤,장봉균,김광섭,김광섭,이학주,최병익,이승모
20 해외특허	중국	투명 그래픽 전극과 이의 제조방법, 및 이를 이용한 슈퍼캐시터	2.0158E+11	2017-09-25	ZL201580076	2019-03-26	조경민	조경민,김광섭,김상민,이승모,김재현,이학주
21 해외특허	중국	탄화비나눔 및 산화비나눔이 장식되어 있는 탄소섬유 및 이의 제조법	2.0188E+11	2020-03-27			이승모	이승모,김재현,이학주,황보윤,김광섭,장봉균,김경식
22 해외특허	중국	유무기 복합체 및 이의 제조방법	2.0188E+11	2020-08-12			이승모	이승모,김재현,김경식,황보윤,김광섭,장봉균,이학주,문명운